

Problem Set 4: Solution

1. Difference equation to $H(z)$

A system obeys $y[n] - 1.4y[n-1] + 0.45y[n-2] = x[n] + x[n-1]$.

- a) Find $H(z)$.
- b) Determine **all** zeros and poles of $H(z)$.
- c) Determine the ROC if the system is causal.
- d) Is the causal system BIBO stable?

$$\begin{aligned} \text{a) } H(z) &= \frac{1 + z^{-1}}{1 - 1.4z^{-1} + 0.45z^{-2}} = \frac{1 + z^{-1}}{(1 - 0.5z^{-1})(1 - 0.9z^{-1})} \\ &= \frac{z(z+1)}{(z-0.5)(z-0.9)} \end{aligned}$$

$$\begin{aligned} \text{b) zeros: } z &= 0, z = -1 \\ \text{poles: } z &= 0.5, z = 0.9 \end{aligned}$$

$$\begin{aligned} \text{c) For causal (right-sided } h[n]) \text{ system} \\ \text{ROC: } |z| &> 0.9. \end{aligned}$$

$$\begin{aligned} \text{d) Yes, the causal system is BIBO stable} \\ \text{since the unit circle is included in the ROC.} \end{aligned}$$

2. Geometric frequency-response reasoning

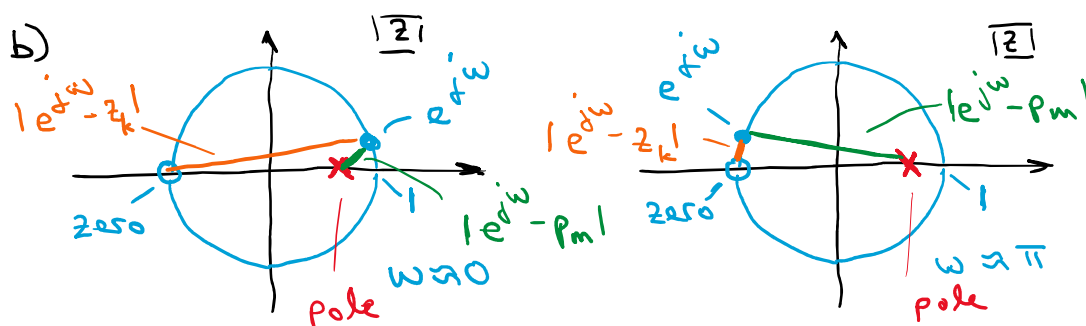
Consider $H(z) = (1 + z^{-1})/(1 - 0.8z^{-1})$.

a) Determine the poles and zeros of $H(z)$.

b) Use a geometric method to compare $|H(e^{j\omega})|$ near $\omega = 0$ and $\omega = \pi$. Show a labeled sketch of the geometric evaluation.

c) What is your conclusion about the qualitative filter type of this $H(z)$?

$$a) H(z) = \frac{1 + z^{-1}}{1 - 0.8z^{-1}} \quad \text{zero: } z = -1, \text{ pole: } z = 0.8$$



$$|H(e^{j\omega})| = \frac{\text{large}}{\text{small}} = \text{large for } \omega \approx 0$$

$$|H(e^{j\omega})| = \frac{\text{small}}{\text{large}} = \text{small for } \omega \approx \pi$$

c) Conclusion for smaller at $\omega \approx \pi$ than at $\omega \approx 0$ is lowpass filter

in fact

$$|H(e^{j\omega})|_{\omega=\pi} = 0$$

3. 600 Hz notch filter design

At $f_s = 9000$ Hz, design the pole-zero locations of a second-order notch at 600 Hz using zeros on the unit circle and poles at radius $r = 0.96$ with the same angles ω_0 .

a) Give ω_0 , the two zero locations and the two pole locations.

b) Explain what increasing r to 0.99 would do.

$$\begin{aligned} \text{a) Nyquist frequency } f_s/2 &= 4500 \text{ Hz} \\ \frac{4500}{600} &= 7.5 \quad \frac{180^\circ}{7.5} = 24^\circ \quad \frac{\pi}{7.5} = 0.419 = \omega_0 \end{aligned}$$

$$\begin{aligned} \text{zeros at } z &= e^{\pm j 2\pi/15} = \cos\left(\frac{2\pi}{15}\right) \pm j \sin\left(\frac{2\pi}{15}\right) \\ &= 0.914 \pm j 0.407 \end{aligned}$$

$$\begin{aligned} \text{poles at } p &= 0.96 e^{\pm j 2\pi/15} = 0.96 \left[\cos\left(\frac{2\pi}{15}\right) \pm j \sin\left(\frac{2\pi}{15}\right) \right] \\ &= 0.877 \pm j 0.390 \end{aligned}$$

b) Increasing r to 0.99 would make the notch sharper and more narrow since the pole would move closer to the unit circle.

Tradeoff: The filter would become more sensitive to small numerical errors and to coefficient quantization.

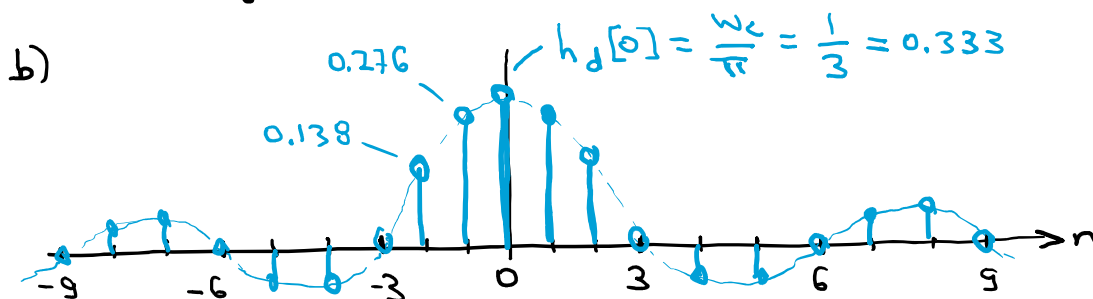
4. Ideal lowpass FIR filter and causal realization

An ideal discrete-time lowpass filter has

$$H_d(e^{j\omega}) = \begin{cases} 1, & |\omega| \leq \omega_c, \\ 0, & \omega_c < |\omega| \leq \pi. \end{cases}$$

- Use the inverse DTFT to determine the unit impulse response $h_d[n]$
- Sketch and label $h_d[n]$ for $\omega_c = \pi/3$ and n in the range $[-8, 8]$. What is the value of $h_d[0]$?
- Explain why this ideal impulse response cannot be implemented exactly as a real-time FIR filter.
- For an odd-length filter with $N = 61$ taps, state the shift required to make a symmetric truncated version causal and give its constant group delay.

$$\begin{aligned} \text{a) } h_d[n] &= \frac{1}{2\pi} \int_{-\omega_c}^{\omega_c} e^{j\omega n} d\omega = \frac{1}{2\pi} \left. \frac{e^{j\omega n}}{jn} \right|_{-\omega_c}^{\omega_c} \\ &= \frac{e^{j\omega_c n} - e^{-j\omega_c n}}{j2\pi n} = \frac{\sin(\omega_c n)}{\pi n} = \frac{\omega_c}{\pi} \text{sinc}\left(\frac{\omega_c n}{\pi}\right) \end{aligned}$$



- c) $h_d[n]$ is non-causal and has infinite duration from $n = -\infty$ to $n = +\infty$. A realizable FIR filter must have a finite number of taps $\Rightarrow$ truncation of $h_d[n]$ is needed for real-time FIR.

- d) If $N=61$ we need $h[n] = h_d[n-30]$, a shift of 30. The constant group delay is $(N-1)/2 = 30$ samples.

5. Window tradeoff

A student designs two 51-tap lowpass filters with the same cutoff. Filter A uses a rectangular window; Filter B uses a Hamming window.

- a) Which filter should have the narrower transition band?
- b) Which filter should have lower stopband sidelobes?
- c) Give approximate first-sidelobe levels for the two windows.
- d) Explain why merely increasing the rectangular-window length does not remove the Gibbs overshoot near a discontinuity.

a) The rectangular window has a narrower transition band

b) The Hamming window has much lower sideband lobes

c) Rectangular window has ≈ -13 dB first-sidelobe level relative to peak level

Hamming window has ≈ -43 dB first-sidelobe level relative to peak level

d) Increasing rectangular window length N brings the overshoot ripples closer together but does not decrease the height of the overshoot. Gibbs phenomenon is a result of the discontinuity of the rectangular window and the $\approx 9\%$ overshoot of the jump size never decreases.

6. Symmetry and linear phase

Let a real-valued FIR filter have length N and satisfy

$$h[n] = h[N - 1 - n].$$

a) Show that its frequency response can be written in the form

$$H(e^{j\omega}) = e^{-j(N-1)/2} A(\omega),$$

where $A(\omega)$ is real-valued.

b) What is the group delay where $A(\omega) \neq 0$? Why can this be useful in machine learning in a feature-extraction pipeline?

a) $n = 0, 1, 2, \dots, N-1$ where N is odd

Shift $h[n]$ left by $\frac{N-1}{2}$ to make it symmetric around $n=0$

$$g[n] = h\left[n + \frac{N-1}{2}\right] \Leftrightarrow G(e^{j\omega}) = H(e^{j\omega}) e^{j\omega \frac{N-1}{2}}$$

$$\begin{aligned} \text{Since } h[n] &= h[N-1-n], \quad h\left[N-1 - \left(n + \frac{N-1}{2}\right)\right] = \\ &= h\left[\frac{N-1}{2} - n\right] = g[-n] \end{aligned}$$

$$\text{Thus, } g[n] = g[-n], \quad n = -\frac{N-1}{2}, \dots, 0, \dots, \frac{N-1}{2}$$

$$\begin{aligned} G(e^{j\omega}) &= \sum_{n=-\frac{N-1}{2}}^{\frac{N-1}{2}} g[n] e^{-j\omega n} = \\ &= \sum_{n=-\frac{N-1}{2}}^{-1} g[n] e^{-j\omega n} + g[0] + \sum_{n=1}^{\frac{N-1}{2}} g[n] e^{-j\omega n} \\ &= g[0] + \sum_{n=1}^{\frac{N-1}{2}} \left(\underbrace{g[-n]}_{=g[n]} e^{j\omega n} + g[n] e^{-j\omega n} \right) \\ &= g[0] + \sum_{n=1}^{\frac{N-1}{2}} g[n] \underbrace{(e^{j\omega n} + e^{-j\omega n})}_{=2\cos(\omega n)} \end{aligned}$$

Thus $G(e^{j\omega}) = A(\omega)$ where

$$A(\omega) = g[0] + 2 \sum_{n=1}^{\frac{N-1}{2}} g[n] \cos(\omega n) \quad \text{is real-valued}$$

$$\text{and } g[n] = h[n + \frac{N-1}{2}], \quad n = 0, 1, \dots, \frac{N-1}{2}$$

$$\text{Since } G(e^{j\omega}) = H(e^{j\omega}) e^{j\omega \frac{N-1}{2}}$$

$$\Rightarrow H(e^{j\omega}) = A(\omega) e^{-j\omega \frac{N-1}{2}}$$

amplitude
term
real-valued

phase term
 $\Theta(\omega) = -\frac{N-1}{2} \omega$

b) group delay: $\tau_g = -\frac{d\Theta(\omega)}{d\omega} = \frac{N-1}{2}$

A constant (with respect to ω) group delay means every frequency component of the input signal is delayed by the same number of samples. In feature extraction processing (e.g. using a filterbank), this guarantees that transients and relative timing relationships that may be important for a ML algorithm stay aligned over all filter channels.

7. Pole radius and time-domain decay (ECEN 5632 Problem)

A causal system has a complex pole pair $p = r e^{\pm j\theta}$. Explain why the corresponding impulse-response terms decay roughly as r^n and why moving r toward 1 simultaneously increases ringing duration and sharpens the frequency-domain resonance. Connect your argument to geometric distance from the unit circle.

$$p = r e^{\pm j\theta} \Rightarrow H(z) = \frac{N(z)}{(1 - r e^{j\theta} z^{-1})(1 - r e^{-j\theta} z^{-1})}$$

$$H(z) = \frac{N(z)}{1 - 2r \cos(\theta) z^{-1} + r^2 z^{-2}}$$

Using z -transform table

$$\frac{1 - r \cos(\theta) z^{-1}}{1 - 2r \cos(\theta) z^{-1} + r^2 z^{-2}}, \quad |z| > r \Leftrightarrow r^n \cos(\theta n) u[n]$$

$$\frac{r \sin(\theta) z^{-1}}{1 - 2r \cos(\theta) z^{-1} + r^2 z^{-2}}, \quad |z| > r \Leftrightarrow r^n \sin(\theta n) u[n]$$

Thus $H(z) \Leftrightarrow h[n] = r^n [\alpha \cos(\theta n) + \beta \sin(\theta n)] u[n]$,
i.e. $h[n]$ is a damped sinusoid (if $r < 1$) with envelope r^n . As r gets closer to 1, r^n takes longer to decay and the frequency domain resonance becomes sharper. Geometrically, the term $|e^{j\omega} - p_k|$ in the denominator of $|H(e^{j\omega})|$ becomes smaller as ω gets close to θ in $p_k = r e^{\pm j\theta}$, resulting in large $|H(e^{j\omega})|$ near $\omega = \pm \theta$.